SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Technology Stack

Date	8 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap	Provides a clean, responsive layout ensuring credit officers can enter data comfortably across mobile field tablets or desktop terminals.
2	Backend / Server-Side	Python, Flask Web Framework	Chosen for its lightweight structure, quick capacity to run native Python packages, and minimal routing setup

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
			needed to wrap machine learning model logic inside clean web endpoints.
3	Database / Data Storage	Serialized Pickle (.pkl), Pandas DataFrames	Efficiently handles temporary structured evaluation records during web app operations and instantly deserializes pre-trained algorithm configurations for real-time predictions.
4	Cloud / Hosting / Deployment	IBM Cloud	Offers a highly scalable, reliable enterprise infrastructure hosting environment optimal for serving Python-based prediction microservices.
5	Version Control & CI/CD	Git, GitHub	Facilitates smooth repository code-tracking, branch management, and robust source collaboration among team members.
6	Third-Party APIs / Other Tools	XGBoost Classifier, Scikit-Learn, NumPy, Pandas	Provides the operational data science pipeline framework to execute statistical preprocessing (mean/mode imputation) and the core predictive model.